AVIATION SYSTEMS ENGINEERING · RESEARCH CONCEPT REPORT

# ANIRBAN-1: IoT Mesh for Autonomous Aircraft Safety

## Integrated Technical Concept Report — Cooperative Trajectory Sharing, Conflict Prediction, and Advisory-Layer Engineering

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| Report date | Revision | Document type | System boundary |
|---|---|---|---|
| 15 October 2026 | 1.0 | Concept / research architecture | Advisory-only, simulation-first |


> **Status: Research Concept — Not for Operational Use.** This document is a research architecture and concept report. It is not an installation manual, a certified design, an operational flight procedure, or evidence of compliance with any standard named in it. Naming a standard does not establish compliance. Any real-aircraft modification must be designed and approved by qualified aviation specialists and the responsible authority.

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## 1 Document Status and Scope of This Report

This is revision 1.0 of the ANIRBAN-1 concept report. It integrates the technical substance of the supplied concept report and its accompanying research-and-technology comparison and system-architecture design into one document, and it revises the presentation of the original concept where the original overstates evidence.

Two source representations were inspected: the report HTML (primary editable source) and its rendered PDF (quality-assurance reference). They are different renderings of the same study, not two separate studies. The PDF inspection identified the defects corrected here: an opening chart gallery detached from the narrative, tables that collapsed into single merged cells and misaligned headers on export, a duplicated glossary and contributor block at the end of the document, substantial blank page area, and a radar update interval drawn as a single fixed value at the top of a reported 4–12 second range.

Every visualization in this edition is hand-authored inline SVG. No remote chart or diagram script is loaded, so figures cannot render blank or malformed during print or PDF export. Sections 2–19 carry the technical content; Section 15 records what was changed relative to the original concept and why.

> **Engineering boundary maintained throughout:** ANIRBAN-1 is a simulation-first, advisory-only research concept. It does not command flight controls, and it does not replace, disable, or contradict existing TCAS/ACAS, ADS-B, ATC, or established operating procedures. No claim is made that the proposed system is safer, more accurate, more reliable, or more effective than existing aviation systems.

## 2 Executive Summary

ANIRBAN-1 proposes a cooperative, advisory-only aircraft safety overlay. Participating aircraft would exchange state and planned-trajectory intent, predict potential conflicts over a declared horizon, evaluate candidate resolutions against explicit constraints, and present an explainable recommendation to a pilot or operator. Existing collision-avoidance and air-traffic functions remain independent and unmodified throughout.

**The founding research question is answerable, and the answer is currently "unproven but testable."** The question — whether a carefully engineered, fault-aware cooperative trajectory-sharing and conflict-prediction layer can demonstrate measurable benefits beyond existing surveillance, collision-avoidance and trajectory-management approaches — can be tested on bench computers and simulators at low cost. However, no public evidence establishes that such a layer already exceeds the ACAS X / DAIDALUS / trajectory-based-operations class of systems. The defensible research frontier is narrower than the original concept assumed: not smarter avoidance logic, and not a new network, but whether aircraft can share intent and coordinate under fault, uncertainty and disagreement without ever making their existing safety depend on that cooperation.

Six corrections to the original concept pitch are carried through every subsequent section:

1. **TCAS is not a last-second panic button.** ACAS II / TCAS II operates independently of ground ATC and issues Traffic and vertical Resolution Advisories. It is a complement ANIRBAN-1 must never contradict.
2. **Ground radar is not obsolete.** ADS-B complements surveillance with roughly one-second GNSS-derived broadcasts, but visibility depends on equipage, coverage and reception, and the broadcast carries no native authentication.
3. **6G is not an available aviation safety network.** IMT-2030 performance requirements were drafted in February 2026 with commercial framing around 2030. A dedicated 6G air-to-air mesh is future research, not a prototype dependency.
4. **Consensus and ledgers cannot establish sensor truth.** Authentication proves message origin and integrity only. General-purpose blockchain consensus is removed from the time-critical loop; coordination uses a deterministic, timeout-bounded proposal/acknowledgement protocol with declared disagreement outcomes.
5. **LiDAR, cameras and radar are not universal detectors.** Airborne small-target detection is a bounded research question requiring measured probability-of-detection and false-alarm characteristics against a declared target envelope.
6. **The mesh itself is not the invention.** IEEE 1920.1 already defines infrastructure-free air-to-air ad hoc networking. ANIRBAN-1 should build on such services and position its contribution above the transport layer.

The recommended Version 0.1 is defined by ten system requirements (SYS-001 to SYS-010, Table 4), two of which are hard gates: no flight-control output (SYS-007) and fail-closed suppression of advice when inputs are inadequate (SYS-006). Validation proceeds through stages V0–V6 with declared exit evidence, benchmarked against three baselines and using ablation runs that disable one feature at a time so that any measured improvement is attributed rather than asserted. Novelty is stated strictly as four hypotheses (C1–C4, Table 22), each paired with the evidence it would require.

Numbers that could not be verified in the supplied sources — per-aircraft cost bands, power draw, fuel-burn impact, communication uptime targets, alert lead-time figures, certification timelines, and assigned assurance levels — are recorded as unverified or removed rather than replaced with invented estimates. Section 15 lists every such decision.

# 3 Introduction, Motivation, Problem Statement, and Research Question

## 3.1 Motivation and industry context

The motivation for a cooperative layer is that tactical separation today is delivered by systems designed for a different problem. Onboard collision avoidance acts close to an encounter and normally constrains vertical manoeuvres; surveillance broadcast publishes current state, not a negotiated future path; and air-traffic services coordinate strategically through ground-mediated processes. A peer-to-peer exchange of intent is the missing element, and the commercial aviation connectivity market is often cited as evidence that the enabling infrastructure is maturing. Figure 1 shows that market framing, and the caveat matters more than the curve.

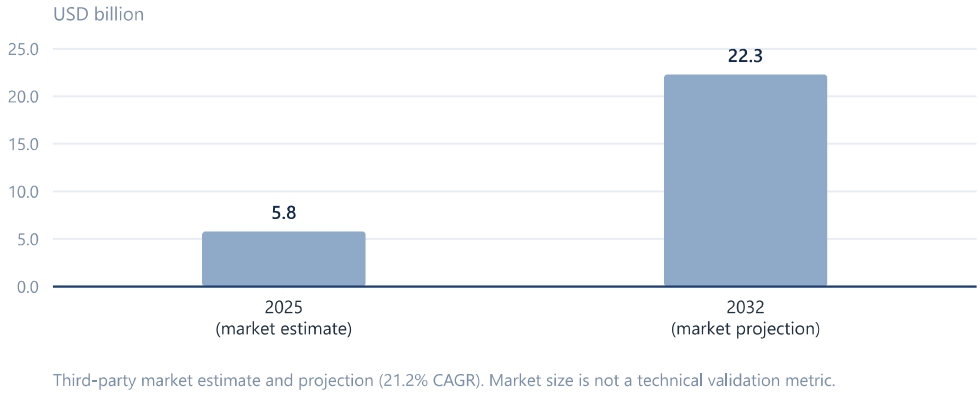



**Figure 1. Global aviation IoT market: reported estimate and projection**

Third-party market research reports the aviation IoT market at USD 5.8 billion in 2025 (estimate) with a projection of USD 22.3 billion by 2032, a compound annual growth rate of 21.2%. Both bars are third-party figures, not measurements, and the two points are separated by seven years of forecast uncertainty.

Source: Market research report cited in the supplied research-and-technology comparison (accessed 15 October 2026). Reference 33 in Section 17.

**Interpretation limit.** Market size is not a technical validation metric. The reported growth concentrates in predictive maintenance, asset tracking and cabin services; the same industry surveys identify cybersecurity, functional safety and sector conservatism as the three adoption challenges. A safety-adjacent collision function sits at the hardest certification tier of exactly those three challenges. Figure 1 therefore supports a statement about connectivity maturity only, and is retained here for context rather than as evidence that ANIRBAN-1 would work.

### 3.2 What existing collision avoidance already achieves

Any claim about an additional safety layer must be measured against the systems that already exist. The most important reference point is the successor to TCAS II. The ACAS X development line replaces fixed rule tables with Markov-decision-process logic and machine-learning-optimized lookup tables, supports vertical, horizontal and blended resolution advisories, and fuses ADS-B In data. Programme-reported results attribute a 65 per cent reduction in unnecessary alerts and a 20 per cent reduction in mid-air collision risk relative to TCAS II, as shown in Figure 2.

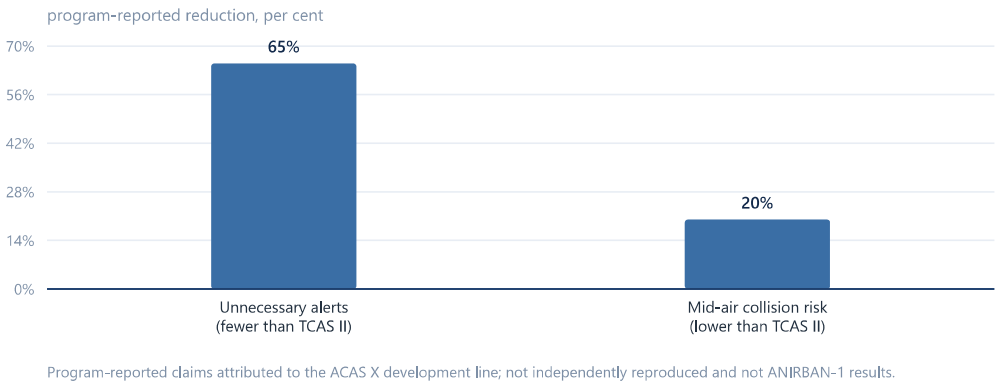



**Figure 2. ACAS X programme-reported improvements relative to TCAS II**

Two programme-reported reductions: 65% fewer unnecessary alerts and 20% lower mid-air collision risk. These are claims attributed to the ACAS X development line. They have not been independently reproduced in this report, and they are not ANIRBAN-1 results.

Source: Programme data summarized in the supplied research-and-technology comparison; corroborated by the ACAS X programme overview. References 3 and 19 in Section 17.

The implication for ANIRBAN-1 is uncomfortable but clarifying. Optimized, sensor-flexible, probabilistic collision avoidance with multi-directional advisories is a funded, standardized, legislatively advancing programme: European legislation already accepts configurations requiring aircraft above defined seat or weight thresholds to equip ACAS Xa or ACAS II 7.1. ANIRBAN-1 novelty therefore cannot rest on smarter avoidance logic. What remains open is the peer-to-peer fault semantics above it — and Figure 2 sets the bar against which any ANIRBAN-1 alerting claim must be compared, not surpassed by assertion.

## 3.3 Related technology: trajectory-based operations

Sharing a future path is not itself unprecedented. Trajectory-based operations (TBO) share and dynamically maintain a four-dimensional trajectory among air traffic control, airlines and aircraft, with the flight management system managing speed against required arrival times. National programmes have moved this from paper to flight. Figure 3 reports the contents of one such verification.




**Figure 3. Reported contents of a two-aircraft trajectory-based-operations verification**

Reported counts from a completed two-aircraft verification: 31 verification subjects, 28 datalink command and information items, and 14 aircraft-to-aircraft and air-ground situation-sharing items, all reported at 100% success. The 100% figure refers to the success of those reported items in that programme.

Source: Civil aviation authority news report of a completed two-aircraft TBO verification (December 2024), cited in the supplied research comparison. Reference 24 in Section 17.

Figure 3 establishes that minutes-horizon trajectory sharing between aircraft and ground services is being implemented by state programmes. It does not establish anything about ANIRBAN-1. A two-aircraft verification does not demonstrate safety case adequacy, scalability to dense traffic, certification readiness, or the effectiveness of a different architecture. Its relevance to this report is narrow and specific: it confirms that the intent-sharing premise rests on live groundwork, which shifts ANIRBAN-1 contribution away from the trajectory mathematics and toward what those programmes do not yet supply — aircraft-to-aircraft intent exchange with declared fault semantics between peers, without waiting for ground-mediated agreement.

### 3.4 Problem statement and research question

The problem ANIRBAN-1 addresses is therefore not the absence of collision avoidance, surveillance or trajectory management. It is the absence of a documented, testable specification for *peer coordination under fault*: what an aircraft does when a peer's intent message is stale, missing, contradictory, expired, or adversarial; what happens when the network partitions; and how a human is told that the system no longer knows enough to advise.

The research question is stated for testing, not for advocacy:

> **Research question.** Can a carefully engineered, fault-aware cooperative trajectory-sharing and conflict-prediction layer demonstrate measurable benefits beyond existing aviation surveillance, collision avoidance, and trajectory-management approaches — where "benefits" are defined by pre-declared metrics, measured against named baselines, and attributed by ablation?

This report answers the question honestly: at concept stage the answer is that the question is testable and the testbed is inexpensive, and that the benefit remains undemonstrated. Sections 12 and 14 define what would constitute an acceptable answer.

## 4 System Scope, Objectives, Assumptions, and Non-Goals

### 4.1 Version 0.1 scope

Version 0.1 operates primarily in simulation. It processes simulated or appropriately available aircraft-state and traffic information; maintains a local representation of nearby traffic with declared uncertainty; predicts potential conflicts over a declared prediction horizon; evaluates alternative trajectories under explicit constraints; exchanges simulated aircraft intent and candidate proposals; handles stale messages, communication loss, disagreement and invalid inputs; presents recommendations through a research interface; and remains isolated from flight-control commands, leaving existing ACAS/TCAS functions and applicable operational procedures independent.

It does not begin by assuming that aircraft automatically execute maneuvers negotiated by an algorithm. Expansion paths — hardware-in-the-loop testing, possible approved aircraft integration, additional sensors, new communication services, eventual certification — are described separately in Section 13, each with its own prerequisites and exit criteria.

**Table 1. Version 0.1 design decisions and rationale**

| Decision area | Baseline choice | Reason |
|---|---|---|

| Decision area | Baseline choice | Reason |
|---|---|---|
| Product boundary | Advisory-only cooperative conflict predictor | Meaningful testing without controlling an aircraft |
| Initial traffic data | Simulated tracks; recorded data where permitted | Repeatable tests before any real aircraft interface |
| Compute | Research workstation / bench computer | Avoids prematurely selecting flight-qualified hardware |
| Network | Virtual network; lab network only under authorization | Tests intent messages, delay, loss and partitions safely |
| Coordination | Deterministic proposal/acknowledgement with timeouts | Simpler to test than consensus; explicit failure behaviour |
| Sensors | Optional module, evaluated separately | Camera/LiDAR/radar performance needs target- and environment-specific evidence |
| Display | Conventional advisory panel first | Human-factors evidence before any AR or HUD integration |
| Aircraft integration | None until aircraft and approval path are defined | Placement, buses, power, antennas and approvals are airframe-specific |

Consolidated from the supplied system-architecture design document (Rev 0.1). These choices replace the original concept report's framing, in which a negotiated trajectory was fed to the flight-control system and the pilot's default was to follow it automatically.

### 4.2 Explicit non-goals

Six non-goals define the evidentiary boundary of Version 0.1. Each removes a burden of proof the programme cannot currently meet, and each is paired with a future expansion path that carries its own prerequisites (Section 13).

- Replace TCAS/ACAS, ADS-B, radar, ATC, the flight management system, or operating procedures.
- Automatically steer, climb, descend, slow, or send commands to an autopilot or flight-control computer.
- Assume all aircraft carry ANIRBAN equipment or share complete future four-dimensional trajectories.
- Claim that LiDAR or cameras see all birds, drones, balloons, gliders, or non-cooperative aircraft.
- Require live 6G, cloud connectivity, or blockchain to detect a local conflict.
- Claim certification compliance on the basis of this concept document.

The second non-goal is decisive. Once the system has no write path to flight control, its worst failure class changes from an aircraft-level safety event to misleading information, which is what permits bench-computer development at research pace. The sixth pre-empts the most common failure mode of concept documents of this kind: presenting a list of standard names as if it were an assurance argument.

### 4.3 Standing assumptions

- **Unequipped traffic exists.** A portion of nearby traffic will never participate; the local picture must remain usable with cooperative data only.

- **Intent is a claim.** A received intended trajectory is a statement about planned behaviour, not a guarantee of future position.
- **Networks fail.** Loss, delay, reordering, duplication and partition are normal conditions to be designed for, not anomalies.
- **Authentication is not truth.** A correctly signed message may still carry an incorrect navigation solution, a wrong clock, or a compromised sender.
- **Access is permission-bound.** Direct access to flight-management or collision-avoidance internals is aircraft-specific and must never be assumed.
- **A criterion must be declared.** Every conflict result depends on the chosen separation or well-clear criterion and its operating domain.

# 5 System Architecture and Requirements

## 5.1 Layered architecture

The architecture separates external data acquisition, local airspace state, prediction, planning and coordination, an independent proposal check, the human interface, and evidence collection. Its organising principle is evidential hygiene: nothing reaches the prediction path without a timestamp, a source identity, a validity flag, and an uncertainty representation or an explicitly declared unknown. Figure 4 shows the layers and their boundaries.

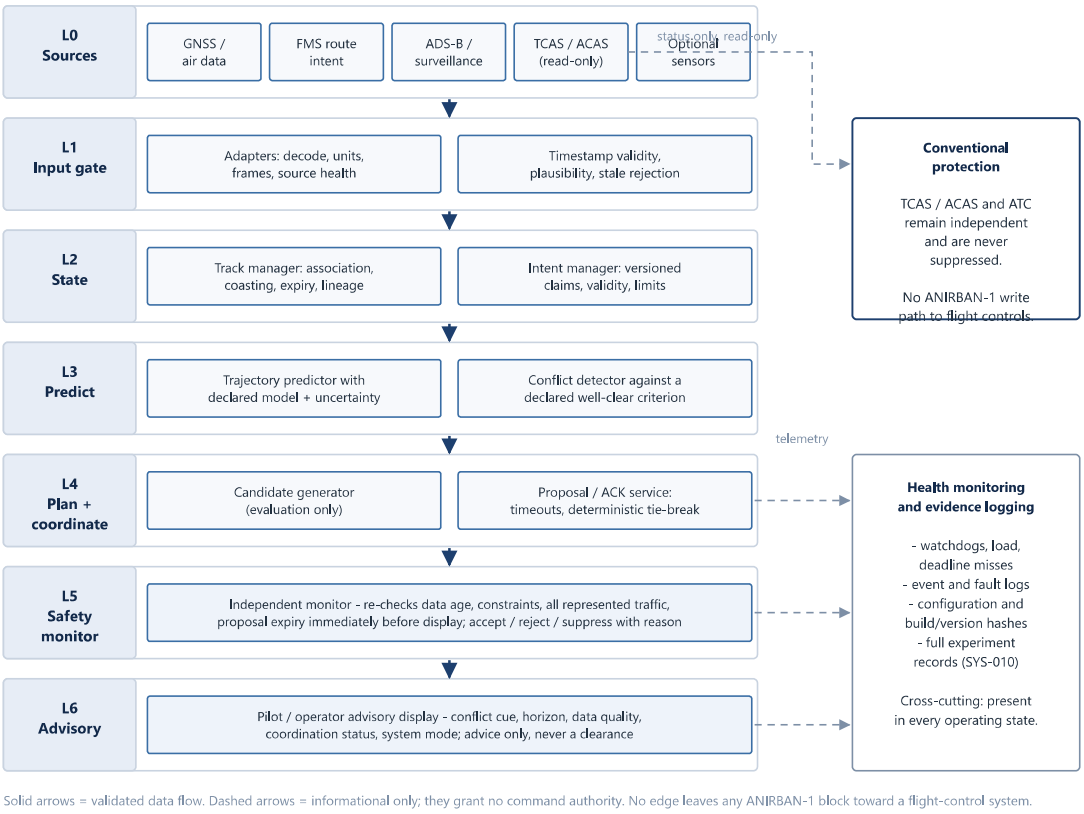



**Figure 4. ANIRBAN-1 Version 0.1 layered system architecture (L0–L6)**

Data flows downward through seven logical layers from aircraft and scenario sources (L0) to the advisory display (L6). The independent safety monitor (L5) re-checks every candidate immediately before display. Conventional protection (TCAS/ACAS and ATC) and the health/evidence function are drawn outside the main data path: dashed links are informational and grant no command authority. No edge leaves any ANIRBAN-1 block toward a flight-control system.

Source: Engineering design consolidated from the supplied system-architecture design document (Rev 0.1, 15 October 2026) and extended for this revision.

Figure 4 illustrates two properties that are architectural rather than stylistic. First, the independent monitor sits between planning and display, so an unchecked candidate cannot reach the pilot by passing through the planner alone. Second, conventional protection is drawn as a separate boundary that the advisory layer may read but never suppress. Together these encode requirement SYS-007 and hazard HZ-06 structurally.

**Table 2. Logical layers L0–L6 and cross-cutting functions**

| Layer | Main components | Primary outputs |
|---|---|---|
| L0 Aircraft / scenario sources | Navigation and air data; route intent where available; traffic surveillance (ADS-B derived); optional sensors; ATC/ACAS state only where properly provided | Raw observations with source, time and quality |
| L1 Input boundary | Source adapters; unit normalization; timestamp checks; validity rules; source health | Validated observation events, or explicit invalid-data states |

| Layer | Main components | Primary outputs |
|---|---|---|
| L2 State estimation | Track manager (create, associate, coast, expire, source lineage); intent manager (versioned claims); ownship state; uncertainty and data age | Versioned local airspace state |
| L3 Prediction | Trajectory model with declared assumptions and uncertainty propagation; conflict detector against a declared criterion | Predicted conflict intervals with reason codes |
| L4 Planning / coordination | Candidate generator; multi-traffic re-check; proposal and acknowledgement exchange; deterministic tie-break and timeout logic | Candidate advisory proposal, or an explicit no-safe-recommendation state |
| L5 Safety monitor | Independent re-check of input validity, constraints, all represented traffic and proposal expiry | Accept, reject or suppress, with reason code |
| L6 Human interface | Advisory display; path visualization; confidence and data-age banner; acknowledgement | Pilot or operator information only |
| Cross-cutting | Health monitoring; logging; cybersecurity; configuration and version control; test harness | Traceable evidence and diagnosis |

## 5.2 Operating states

Layer L5 and the advisory manager together enforce a small, closed set of operating states. The requirement is that degradation is always visible and always named: an aircraft's safety-relevant awareness moves through declared states rather than silently losing confidence. Figure 5 gives the state machine; Table 3 defines its transitions.

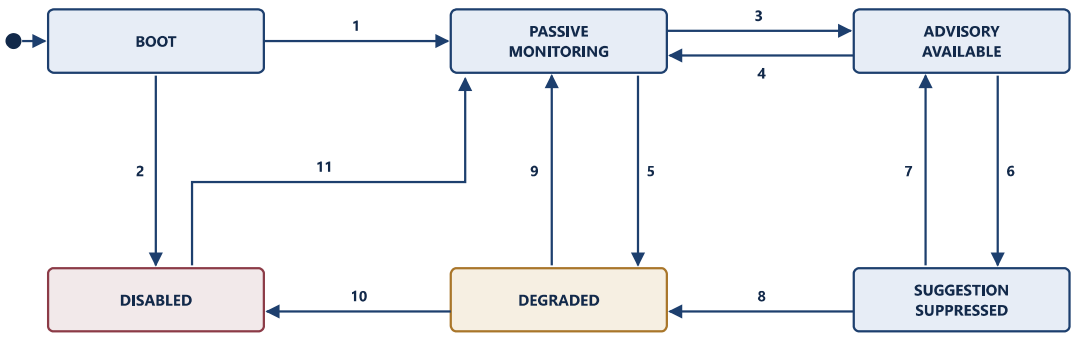



**Figure 5. Operating-state machine for the advisory-only prototype**

Six declared states: BOOT, PASSIVE_MONITORING, ADVISORY_AVAILABLE, DEGRADED, SUGGESTION_SUPPRESSED and DISABLED. Eleven numbered transitions govern entry and exit. Transitions into DEGRADED and DISABLED are triggered by input or timing faults; SUGGESTION_SUPPRESSED is entered when the independent monitor rejects a candidate or a proposal expires.

Source: Derived from the supplied system-architecture design document (Rev 0.1) and the failure-handling table reproduced as Table 3.

**Table 3. State transitions and triggers**

| # | From | To | Trigger | Required logging |
|---|---|---|---|---|

| # | From | To | Trigger | Required logging |
|---|---|---|---|---|
| 1 | BOOT | PASSIVE_MONITORING | Self-test passed and inputs valid | Build/config hash, self-test result |
| 2 | BOOT | DISABLED | Self-test failed or unauthorized configuration | Failure reason, configuration identity |
| 3 | PASSIVE_MONITORING | ADVISORY_AVAILABLE | Conflict predicted and candidate accepted by the monitor | Conflict set, candidate identifier, monitor decision |
| 4 | ADVISORY_AVAILABLE | PASSIVE_MONITORING | Conflict resolved below the declared criterion | Resolution time, criterion tag |
| 5 | PASSIVE_MONITORING | DEGRADED | Stale source, timing loss or deadline miss | Age, threshold from config, affected outputs |
| 6 | ADVISORY_AVAILABLE | SUGGESTION_SUPPRESSED | Monitor rejects the candidate or the proposal expires | Failed checks, rule identifiers, expiry |
| 7 | SUGGESTION_SUPPRESSED | ADVISORY_AVAILABLE | A fresh candidate is accepted | New candidate identifier, monitor decision |
| 8 | SUGGESTION_SUPPRESSED | DEGRADED | Essential inputs become inadequate | Input status, suppression reason |
| 9 | ADVISORY_AVAILABLE | DEGRADED | Peer unconfirmed where policy requires visibility | Timeout, peer identity, retry count |
| 10 | DEGRADED | DISABLED | Persistent fault or operator disable | Fault latch, operator action |
| 11 | DEGRADED / DISABLED | PASSIVE_MONITORING | Inputs recovered and revalidated, or manual re-enable with successful self-test | Recovery event, revalidation evidence |

Every coordination outcome resolves to one of these states. No transition may depend silently on reaching a peer aircraft; local conflict detection and existing aircraft safety functions remain available in every state.

## 5.3 System requirements

Ten system-level requirements convert the concept into a testable specification. Each carries declared verification evidence, so that requirement satisfaction is an inspectable fact rather than an intention.

**Table 4. Starter system requirements SYS-001 to SYS-010**

| ID | Requirement statement | Verification evidence |
|---|---|---|
| SYS-001 | Accept timestamped ownship and traffic state records from a simulator or authorized data adapter | Interface tests; malformed-data tests |
| SYS-002 | Every track carries source, age, quality status and an uncertainty representation or a declared unknown status | Unit tests; state-store inspection |

| ID | Requirement statement | Verification evidence |
|---|---|---|
| SYS-003 | The predictor produces future-position estimates over a configurable study horizon | Ground-truth simulation comparison |
| SYS-004 | The conflict detector evaluates tracks against a declared criterion selected for the study | Test vectors; baseline comparison |
| SYS-005 | Every proposed alternative is checked against all represented traffic and stated constraints before display | Scenario tests; infeasible-candidate rejection |
| SYS-006 | No valid recommendation is shown if essential inputs are stale, inconsistent or insufficient | Fault-injection tests; user-interface state tests |
| SYS-007 | The prototype outputs no flight-control commands and does not modify TCAS/ACAS behaviour | Interface audit; code and architecture review |
| SYS-008 | Loss, delay, duplication, replay-like input and network partition are observable and logged | Network fault-injection report |
| SYS-009 | Each advisory states the conflict cue, prediction horizon, data-quality indicator, and candidate check status | Human-factors review |
| SYS-010 | Every experiment records scenario configuration, algorithm version, inputs, outputs and metrics | Reproducibility audit |

Retained unchanged from the supplied architecture document. SYS-007 must be verified by architecture and code audit so that the absence of a control path is structurally provable, not merely unexercised in tests. SYS-006 enforces fail-closed behaviour: an inadequate traffic picture produces a suppressed advisory and a visible degraded indication, never a confidently rendered path built from stale data.

## 5.4 Hardware architecture options

Three hardware architectures were compared. Cost and complexity are declared as qualitative categories; no equipment prices are stated, because the supplied documents contain no verified quotations.

**Table 5. Hardware architecture options**

| Option | Composition | Cost / complexity | Stage | Prerequisites to enter the stage |
|---|---|---|---|---|
| H1 — Single research computer | One bench workstation running all modules in separated processes; simulated I/O; browser advisory panel | Low | Phase 1 (simulation MVP) — **recommended start** | None beyond software |
| H2 — Modular research island | Compute host plus isolated input-adapter unit, separate monitoring and watchdog unit, logging store; optional lab radios under authorization | Medium | Phase 3 (HIL prototype) | Stable module interfaces; timing budget measured in H1; authorization for any radio-frequency transmission |

| Option | Composition | Cost / complexity | Stage | Prerequisites to enter the stage |
|---|---|---|---|---|
| H3 — Fault-tolerant airborne-class | Dual cross-monitored processing channels, qualified power and cooling provisions, EMC-hardened enclosures, approved interface gateway, airframe-specific mounting | High | Post-feasibility-study only (Phase 4 and beyond) | Selected aircraft and intended function; authority dialogue; certification basis; ARP4754-class development process established |

H1 is the recommended starting point. Its one substantive hardware discipline is process isolation from the outset: the independent monitor runs as a separate process with its own watchdog, so the assurance argument being built in software anticipates the physical separation that H2 and H3 would require.

**Interfaces, and what is deliberately not specified.** ARINC 429 and ARINC 664/AFDX are plausible future carriers on specific aircraft types, but neither is a universal plug-in interface. Any Interface Control Document must define data definitions, units, label or message sets, update rates, timestamps, quality flags, access rights and failure behaviour before hardware decisions are taken. This report does not specify connector pinouts, antenna placement, bus loading, wire gauges, power budgets, cooling flows or part numbers. The original concept report's installation narrative — top and bottom fuselage antennas, nose and belly sensor fairings with heated windows, avionics-bay compute with cabin ventilation — is demoted to illustrative description, because those are exactly the airframe-specific decisions reserved for a manufacturer-authorized design study. A laboratory prototype and a qualified airborne installation remain distinct categories.

**Table 6. Hardware element register: purpose, prototype status, and open airborne-design questions**

| Element | Purpose | Version 0.1 implementation | Open questions for any airborne installation |
|---|---|---|---|
| Compute host | Runs adapters, track store, prediction, monitor and interface service | Bench computer | Environmental qualification, power draw, cooling, partitioning, worst-case timing, maintainability |
| Input adapter / isolation | Converts simulator or authorized source records into a stable internal schema | Mock interfaces, files, sockets | Permitted aircraft interface, electrical isolation, bus loading, data-access authority |
| Communication gateway | Serializes, validates, authenticates where applicable, expires and routes research messages | Virtual network | Approved spectrum, airworthiness, interference assessment, range, network service, certification evidence |
| Optional sensor module | Turns recorded or simulated detections into bounded target tracks | Off, or recorded-data input | Target envelope, mounting, field of view, environment, EMC, aerodynamic and maintenance impact |
| Operator display | Shows advisory state, confidence, reason and recommended path | Browser or desktop panel | Cockpit integration, alert priority, visibility, latency, workload, training, approval |
| Health and storage | Collects state transitions, inputs, algorithm versions and test outputs | Local files and test database | Secure maintenance access, capacity, retention, privacy and security |

| Element | Purpose | Version 0.1 implementation | Open questions for any airborne installation |
|---|---|---|---|
| Power, cooling and mechanics | Supports all installed equipment | Bench power supply | Load analysis, circuit protection, heat, vibration, fire zones, structural mounting, lightning and EMI |

Fault containment follows the isolated-equipment-island principle: a malfunctioning experimental unit must not affect power quality, bus traffic, or display channels used by certified systems.

# 6 Surveillance Inputs, Sensors, and Non-Cooperative Object Detection

## 6.1 Cooperative surveillance: what ADS-B provides and what it does not

ADS-B broadcasts GNSS-derived aircraft state automatically. ADS-B Out transmits position, velocity and identity about once per second over 1090ES or 978 MHz UAT; ADS-B In receives traffic from other aircraft and ground services, adding traffic information, flight information and relay services. Compared with conventional secondary surveillance radar, reported figures give ADS-B a 0.5-second update interval against a 4–12 second radar band, and roughly 30 metres of position accuracy against roughly 400 metres. Figures 6 and 7 present those two comparisons.

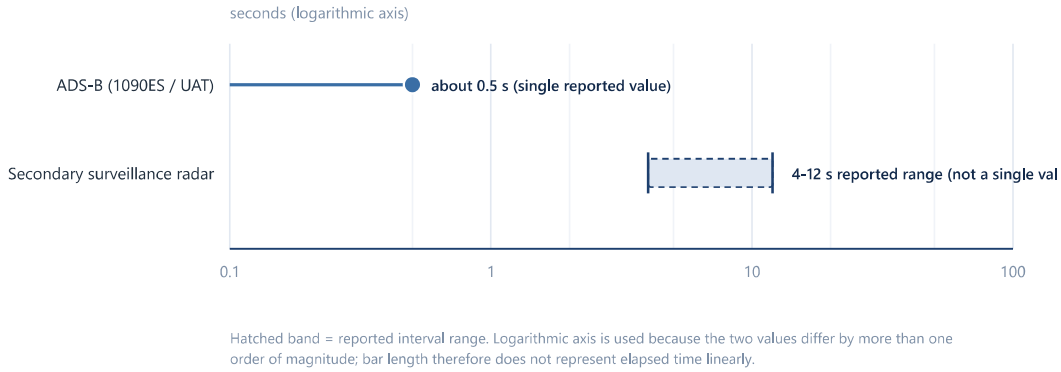



**Figure 6. Reported surveillance update intervals: ADS-B and secondary radar**

ADS-B is reported at about 0.5 seconds (a single reported value, plotted as a point). Secondary surveillance radar is reported as a 4 to 12 second range and is drawn as a hatched band spanning that range, not as a single value. The horizontal axis is logarithmic because the two quantities differ by more than one order of magnitude; bar and band length therefore does not represent elapsed time linearly.

Source: Reported figures cited in the supplied research-and-technology comparison and in civil aviation authority technical material. References 2 and 55 in Section 17.

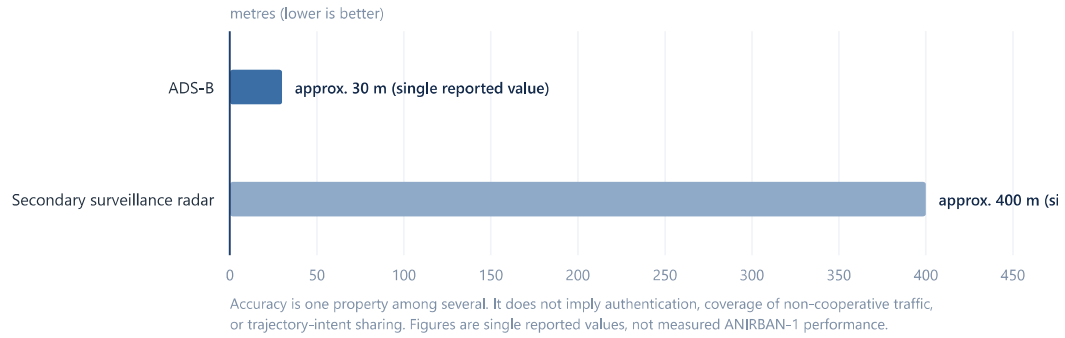



**Figure 7. Reported position accuracy: ADS-B and secondary radar**

Reported position accuracy of approximately 30 metres for ADS-B and approximately 400 metres for secondary surveillance radar. Both are single reported values. Lower is better.

Source: Same sources as Figure 6. References 2 and 55 in Section 17.

Figures 6 and 7 can establish that ADS-B offers more frequent and more precise cooperative state than conventional secondary radar. They cannot establish anything about ANIRBAN-1 performance. Four properties are frequently conflated and are kept separate throughout this report:

- **Update frequency** — how often state is refreshed (Figure 6).
- **Positional accuracy** — how close a reported position is to truth (Figure 7).
- **Authentication and integrity** — whether the receiver can establish origin and immutability. ADS-B was designed without security mechanisms and is susceptible to message injection and sequencing manipulation; neither figure addresses this at all.
- **Traffic coverage and intent** — ADS-B sees transponder-equipped aircraft only, and it publishes current state, not a negotiated future four-dimensional trajectory. This is precisely the gap the ANIRBAN-1 intent layer proposes to fill.

ANIRBAN-1 therefore inherits both the coverage boundary and the integrity gap of its cooperative feed. Every ADS-B-derived track enters the system with age, source and quality flags (SYS-002) and is treated as a time-bounded claim to be cross-checked, not as ground truth.

## 6.2 Non-cooperative sensing: four problems, not one

The supplied concept documents are internally inconsistent about sensor performance, and the inconsistency is recorded here rather than averaged away. The concept body describes a forward-scanning LiDAR at 100–300 metres, depth cameras effective to roughly 1–2 km in clear conditions, and small-target detection radars at roughly 5–10 km; the comparison table at the end of the same document lists LiDAR at 0.5–1 km and cameras at 1–5 km. These ranges differ by multiples within one document. They are carried as unverified design assumptions requiring measurement, not as specifications.

**Table 7. Sensing modality analysis: what each modality observes and what limits it**

| Modality | What it observes | Physical and environmental limits | Integration burden | Version 0.1 role |
|---|---|---|---|---|

| Modality | What it observes | Physical and environmental limits | Integration burden | Version 0.1 role |
|---|---|---|---|---|
| LiDAR | Three-dimensional range and local geometry as a point cloud | Rain and fog attenuate laser return; narrow field of regard; line of sight required; range figures inconsistent across sources | Several kilograms; tens of watts; heated external windows | Recorded-data research module |
| EO / IR camera | Pixel-level imagery and apparent motion | Requires daylight or active infrared illumination at night; degraded by fog, rain, clutter and low contrast | Roughly 1–2 kg; high data rate requiring GPU-class processing | Recorded-data research module |
| Airborne radar | Range, radial velocity and coarse azimuth | Coarse angular resolution; clutter and sidelobe effects; the declared range bounds what can be assured | Tens of kilograms; rotating or phased-array aperture; radome integration | Case-specific option |
| Cooperative surveillance (ADS-B, Mode S) | Broadcast identity and GNSS-derived state | Transponder-equipped traffic only; no native authentication | Existing certified equipment | Baseline input |
| DAIDALUS-style logic | Well-clear prediction derived from tracks | Entirely dependent on input track quality and the declared criterion | Software only | Benchmark, not a sensor |

The detect-and-avoid literature is explicit that no single modality covers the threat set, and that fusion does not automatically produce reliable detection. ACAS Xu was designed to accept flexible sensor inputs including radar so that advice can degrade gracefully when non-cooperative sensing is absent; DAIDALUS supports multi-sensor input through configuration rather than claiming universal coverage.

Detection, identification, tracking and threat assessment are four distinct problems with four distinct error distributions. Published human-in-the-loop research makes the gap concrete: pilots lost well-clear against non-cooperative intruders significantly more often than against cooperative ones, driven by the radar's declared range and sensor noise — that is, by the distance between an instantaneous detection and a usable track with bounded uncertainty. The same body of work records that the format of an alert changes pilot response behaviour and requires dedicated training. ANIRBAN-1 therefore maintains vocabulary discipline: a LiDAR point is not a track; a track is not an identification; an identification is not a threat assessment; and a threat assessment is not a validated avoidance recommendation.

### 6.3 Recommended initial sensing configuration

The smallest sensible initial configuration keeps sensing entirely out of the live loop. The optional sensor module consumes recorded imagery or simulated targets, produces bounded target tracks with explicit confidence, and reports a measured detection and false-alarm distribution against a declared target envelope defined by size class, closure rate, aspect and weather categories.

Features postponed until a separate study justifies them: airborne LiDAR hardware selection; nose or belly fairing mounting with de-icing provisions, which are airframe-specific engineering decisions rather than concept-level ones; weather-radar-adjacent modes; and any universal detection claim. This report states plainly that nothing in the current evidence base supports asserting that a sensor suite detects every bird, balloon, drone, glider or non-cooperative aircraft, and ANIRBAN-1 makes no such claim. Published assessments also characterize the certification basis for non-cooperative sensors as a current gap, which limits beyond-visual-line-of-sight approval in mixed airspace.

# 7 Communications, Message Formats, Coordination, and Failure Behaviour

## 7.1 Link options compared

The original concept treated an aircraft mesh as an open frontier and a dedicated 6G network as an eventual substrate. The evidence requires both framings to be revised: IEEE 1920.1 defines infrastructure-free air-to-air ad hoc networking for civil and commercial manned and unmanned aircraft, with a network services architecture, a security framework and data models, published as a draft standard in October 2022. The design consequence is to stop inventing a transport layer and to place the research contribution above it.

**Table 8. Communication options compared against the properties that matter for safety messaging**

| Option | Coverage | Delay and delay variation | Loss behaviour | Security | Maturity for aviation safety |
|---|---|---|---|---|---|
| Simulated / virtual network | Full control by design | Configurable | Injected deliberately | Application-level | **Chosen for Version 0.1** |
| IEEE 1920.1-class aerial ad hoc networking | Air-to-air, infrastructure-free | Multi-hop dependent | Topology churn dominates | Security framework defined in the standard | Draft standard (October 2022) |
| Existing links: ADS-B extensions, VDL-2 / CPDLC, LDACS, satcom | Operational footprints | 0.5–1 s class | Engineered and regulated | Varies; LDACS designs include national cryptographic protection with TESLA-class control-data authentication | Interim field path; requires standards work |
| Laboratory software-defined radios | Short-range line-of-sight tests | Configurable | Measurable | Application-level | Only under authorization |
| Dedicated 6G aviation mesh | IMT-2030 study phase; three-dimensional coverage targets described from 120 m to 3,000 m | Targeted, not demonstrated | Study phase | Aviation profile undefined | **Future research, not a prerequisite** |

No custom transmission on aeronautical frequencies is prescribed or authorized by this report. Spectrum and authorization constraints are why the virtual-network-first staging is retained.

The safety lesson from airborne-network research is consistent across three strands of published work: backbone-style architectures that keep a connected three-dimensional corridor; aviation ad hoc networks supporting direct, relay and opportunistic transmission to extend air-to-ground access into oceanic and remote airspace; and connectivity-aware mobility models that trade area coverage against network connectivity. With nodes moving at hundreds of kilometres per hour, **link intermittency and message age — not bandwidth — are the governing constraints**. Message age is therefore a first-class protocol field, every peer interaction has a declared deadline, and the safety logic remains functional when the multi-hop path fails.

## 7.2 Conceptual message schema

Five message types carry the safety-relevant exchange. Freshness, versioning and provenance are protocol fields, not afterthoughts.

**Table 9. Conceptual message types, required fields, and semantic rules**

| Message | Required conceptual fields | Rules |
|---|---|---|
| STATE | protocol_version; sender_id; sequence; generated_at; expires_at; position; altitude; velocity; track or heading; vertical rate; uncertainty; source_quality; integrity and authentication metadata | Must be time-stamped, expiring, replay-aware, versioned, unit-defined and quality-marked |
| INTENT | sender_id; intent_version; valid_from; valid_until; path segments or waypoints; expected speed and altitude profile where provided; uncertainty or confidence; source | Intent is a claim about planned behaviour, not a guarantee. Revisions invalidate older versions |
| PROPOSAL | proposal_id; participants and tracks; candidate identifier; intended trajectory summary; assumptions; created_at; expiry; safety-check status; version | A proposal is not a command and not a clearance |
| ACK / REJECT | proposal_id; sender_id; outcome; reason; timestamp; expiry | Missing, late or conflicting acknowledgement means no confirmed agreement |
| HEALTH | component status; degraded reason; link state; current software and configuration version | Diagnostic only; non-critical telemetry must never block the prediction loop |

Encoding choice (a compact binary serialization such as protobuf or ASN.1) is deferred to the prototype; neither choice alters the semantics above.

> **What the message format does not guarantee.** The integrity and authentication metadata field, when populated with a working scheme, establishes message origin and in-flight immutability. It does not establish that the sender's navigation solution is honest, that its clock is synchronized, or that its declared intent will be flown. The schema's answer to that limit is the uncertainty and source-quality fields combined with expiry: every receiver treats message content as a time-bounded claim subject to plausibility and cross-source checks. Authentication and message integrity must not be confused with verification that the underlying sensor information is truthful.

## 7.3 Coordination: deterministic negotiation, not consensus

The original concept invoked a blockchain-inspired consensus protocol to agree conflict-free plans atomically. That design is removed from the time-critical loop for reasons of latency, failure behaviour and, above all, evidential value.

**Table 10. Coordination mechanisms compared for the conflict-resolution loop**

| Criterion | Timeout-bounded proposal / ACK (adopted) | Distributed consensus (Raft or PBFT class) | General-purpose blockchain ledger |
|---|---|---|---|
| Latency behaviour | Bounded by a declared deadline; the cycle continues regardless | Round-dependent; leader or partition delays unbounded in the worst case | Commit times incompatible with a seconds-scale safety cycle |

| Criterion | Timeout-bounded proposal / ACK (adopted) | Distributed consensus (Raft or PBFT class) | General-purpose blockchain ledger |
|---|---|---|---|
| Outcome under peer disagreement | Deterministic tie-break rule, or suppression — a declared state | Consensus may stall; livelock handling is the hard case | Same stall risk, plus non-deterministic fork resolution |
| Behaviour under partition | Local detection and advisory continue; the proposal is marked unconfirmed | Majority quorum may be unreachable precisely when topology is churning | Same, with added replay and ordering exposure |
| What it proves about traffic truth | Nothing beyond message-level controls; plausibility checks remain required | Quorum agreement on received messages, not sensor truth | Record integrity of the ledger, explicitly not source truth |
| Verification burden | Small state machine; every transition testable by fault injection | Formal treatments exist but are heavyweight for a research prototype | Assurance argument effectively unavailable at safety relevance |
| Research value | The fault semantics *are* the candidate contribution | No marginal value over the adopted option for local conflict work | None demonstrated for time-critical collision avoidance |

A ledger retains a legitimate role in ground-side air-traffic-management record integrity, as explored in the literature. It has no role in the airborne detection loop of Version 0.1.

### 7.4 Loss, partition, disagreement and degradation

No aircraft may hold its local conflict detection, its existing safety functions, or its degraded-mode display in reserve behind a global-consensus step. Coordination is enrichment, never a dependency. Resilience design starts from published network behaviour rather than optimism: predictive tooling can forecast data-link interruption minutes ahead, which is useful for strategic deconfliction and useless for an encounter that develops inside a minute; and complex-network vulnerability analysis of aviation systems shows that targeted attacks degrade such networks far more severely than random failures, meaning an intentional adversary will seek the connectivity-critical nodes — relay positions, timing sources and certificate services. Table 11 states the required behaviour.

**Table 11. Failure and degradation behaviour: condition, required response, and evidence**

| Condition | Expected prototype response | Logging |
|---|---|---|
| Source invalid or out of range | Reject the datum; flag the source as degraded; update track status | Source, rule, timestamp and rejection reason |
| Track stale or expired | Mark stale; never treat the last position as current; suppress if the remaining picture is inadequate | Age, thresholds from configuration, affected outputs |
| Intent version changes | Invalidate the older intent; recompute the trajectory and any proposal depending on it | Old and new version, impacted candidate |
| No network response | Continue local prediction; mark any distributed plan unconfirmed; never assume acceptance | Timeout, peer identity, retry count, degraded state |
| Two proposals conflict | Apply the documented deterministic tie-break rule, or suppress the recommendation | Conflict set, rule applied, selected or none result |

| Condition | Expected prototype response | Logging |
|---|---|---|
| Safety monitor rejects a candidate | Not shown as recommended; rejection may be made visible without implying safety | Failed checks and rule identifiers |
| Compute overload or deadline miss | Enter the degraded state; suppress time-expired output; watchdog-controlled recovery | Timing, component health and recovery event |
| Active TCAS/ACAS resolution advisory reported through a permitted input | Issue no contradictory ANIRBAN manoeuvre recommendation; apply the defined caution and mode policy | Input status, suppression reason, version |

The invariant tying Sections 7.2 to 7.4 together: an aircraft's safety-relevant awareness must degrade along declared, observable states, and no transition may silently depend on reaching peers. This is what reduced dependency on continuous connectivity means once stripped of optimistic phrasing — the local loop is self-sufficient and the network is optional enrichment.

# 8 Onboard Software Modules and Processing Workflow

## 8.1 Modules, responsibilities and degraded behaviour

Eleven modules define the software boundary. The discipline that matters is that every failure mode produces an explicit state rather than silence.

**Table 12. Software modules M1–M11: responsibilities and failure behaviour**

| Module | Responsibilities | Failure / degraded behaviour |
|---|---|---|
| M1 Source adapters | Decode simulator or authorized inputs; map units and coordinate frames | Reject unknown schema, impossible values, invalid status or unsupported units |
| M2 Validation and time | Check age, ordering, source status, sequence, clock quality and plausibility | Mark data invalid or stale; never silently reuse old state as current |
| M3 Track manager | Create, update, coast and expire tracks; associate duplicate observations; preserve source lineage | Tracks become degraded or expire; association ambiguity lowers confidence |
| M4 Intent manager | Store trajectory segments with version, sender, start time, expiry and stated limitations | Missing intent triggers an explicitly declared fallback model, not an inferred exact future path |
| M5 Prediction engine | Propagate candidate future positions with model assumptions and uncertainty | Widen uncertainty or report unknown when model or input quality is inadequate |
| M6 Conflict detector | Evaluate predicted trajectories against the configured conflict or well-clear criterion | Return detection state, time window and contributing tracks |
| M7 Candidate generator | Generate bounded alternative paths for evaluation; never command execution | No candidate means the system shows no reliable recommendation |
| M8 Coordination service | Exchange state, intent and proposal messages; correlate acknowledgements; apply timeout and deterministic tie-break policy | No complete agreement by deadline means the proposal is unconfirmed; peer acceptance is never assumed |
| M9 Independent proposal monitor | Re-check data age, constraints, all known traffic, proposal expiry and contradictory state | Reject or suppress the candidate with a clear reason |

| Module | Responsibilities | Failure / degraded behaviour |
|---|---|---|
| M10 Advisory manager | Choose display mode; suppress stale advice; expose reason, quality and time horizon | Transition to a degraded or unavailable display state |
| M11 Health and logging | Watchdogs, processor load, dropped frames, exceptions, event logs, build and configuration hashes | Fault latched and visible; controlled restart where appropriate |

## 8.2 Data structures

Every structure is versioned, and the message-schema evolution rules of Section 7.2 apply at the external boundary. Undocumented ad-hoc object passing between modules is prohibited: the schema is the interface.

**Table 13. Core data structures and their producers**

| Structure | Key fields | Producer | Consumers |
|---|---|---|---|
| ObservationEvent | source_id, rx_time, schema_version, payload, quality_flag | M1 adapters | M2 validator |
| Track | track_id, reduced state vector, covariance or uncertainty, age, source lineage, lifecycle state (fresh, coasting, expired), association set | M3 track manager | M5, M6, M9 |
| IntentRecord | sender_id, intent_version, valid_from and valid_until, segment list, stated assumptions, confidence | M4 intent manager | M5, M9 |
| TrajectoryHypothesis | track_id, model tag, time-indexed states, uncertainty envelope | M5 predictor | M6, M7 |
| ConflictSet | conflict_id, member tracks, interval, criterion tag, reason codes | M6 detector | M7, M9, M10 |
| CandidateProposal | proposal_id, ownship track, adjustment set, constraint report, deviation cost, expiry, safety status | M7 generator | M8, M9, M10 |
| AdvisoryState | mode (monitoring, advisory, degraded, unavailable), displayed content, quality banner, coordination status | M10 advisory manager | Display, M11 log |

## 8.3 Deterministic core with machine learning at arm's length

Four technology routes were compared before selection. Deterministic algorithms — constant-velocity extrapolation and geometric separation tests — are predictable and easy to test, but under-report uncertainty and miss manoeuvring traffic. Optimization methods, such as polynomial trajectory families minimizing deviation cost, produce explicit trade-off logic and reproducible optima but grow combinatorially with participant count. Rule and lookup approaches demonstrate that even learned policy content can be frozen into auditable artifacts at deployment: DAIDALUS-style guidance bands are formally specified, and ACAS X uses machine-learning-optimized numerical lookup tables evaluated offline. End-to-end learning, such as deep reinforcement learning for manoeuvre generation, adapts to uncertainty patterns, but current verification methods for neural components do not meet the predictability required of a safety-adjacent gate.

The adopted architecture therefore uses a deterministic core that owns parsing, units, timeouts, state transitions, conflict thresholds and output gating. A separately monitored machine-learning inference service may contribute estimates — trajectory priors, detection confidence — only through explicit

interfaces and always behind deterministic checks. Technology was selected for verification tractability, not for novelty appeal.

### 8.4 Prediction-to-advisory workflow

The reproducible core loop is the artefact that makes the research question testable. Note the two rejection paths: the monitor gate and the explicit no-reliable-recommendation state.

```
loop every cycle T (configurable; study default 1 s):
  events   = M1.read_raw()                        # timestamped source records
  valid    = M2.filter(events, age_rule, schema_rule, plausibility_rule)
  tracks   = M3.update(valid)                     # create / associate / coast / expire
  intents  = M4.current(versioned)                # declared intent, or FALLBACK tag
  hypo     = M5.predict(tracks, intents, horizon H, uncertainty_model)
  conflict = M6.detect(hypo, criterion = C)
  if conflict:
      cands = M7.generate(conflict, constraints, ownship_envelope)
      cands = M7.evaluate(cands, tracks.all())    # ALL represented traffic, not just th
      coord = M8.propose(cands, peers, deadline D)   # ACK / REJECT / NO-RESPONSE
      for c in cands:
          gate = M9.recheck(c, tracks.age(), C, coord.status)   # independent process
          if gate == ACCEPT: M10.show(c, quality = coord.status)
          else:              M10.suppress(c, reason = gate)
      if no candidate accepted:
          M10.show(NO_RELIABLE_RECOMMENDATION)    # never the least-bad unsafe option
  else:
      M10.show(MONITORING)
  M11.log(cycle_inputs, versions, outputs, timings)   # reproducibility record (SYS-010)
```

**Illustrative labelled simulation example**

The following is a labelled simulation intended to clarify mechanism. It is not performance evidence and asserts no lead-time capability. Two aircraft on a crossing geometry approach a common waypoint; ownship declares a five-minute intent horizon. A constant-velocity baseline flags the conflict using an instantaneous proximity test. The intent-aware variant, propagating both aircraft's speed management against required arrival times, flags earlier but with a widening uncertainty envelope. When the peer's acknowledgement times out at deadline D, the monitor re-checks the candidate against the now-coasting track, finds that the peer's uncertainty is no longer bounded by its declared intent, and suppresses the previously accepted proposal. The display transitions from advisory to degraded with a stated reason. That transition — not a lead-time number — is the research output the scenario demonstrates.

## 9 Human Factors and the Advisory Interface

### 9.1 Design question

The design question is not whether a glowing predicted path looks impressive, but whether a crew can determine at a glance what is measured, what is predicted, what is proposed, and what is broken. Published human-factors evidence constrains the answer. Cockpit traffic display interface requirements

already exist as a standardized human-interface specification covering visual and aural alert characteristics and traffic display requirements. Alerting research shows measurable behavioural trade-offs: as false-alarm rates rise, pilot compliance with alerts falls, while concurrent-task performance can improve because fewer interruptions occur. Three-tier alerting showed no significant advantage over two-tier designs. Automation-assisted flying produced lower situational awareness than manual or interactive modes. Research on the ACAS X family found that the dimension and presentation of an advisory affect pilot response time, and that differing advisory formats require dedicated training. For remote operations, colour and motion cues together with labelled speed data reduce time-to-contact judgement bias, and optimized head-up display presentation reallocates attention usefully in low-visibility conditions.

## 9.2 Advisory display requirements for Version 0.1

**Table 14. Advisory display elements: what must be visible and what must be avoided**

| Display element | Must be visible | Avoid |
|---|---|---|
| Conflict cue | Which tracks are involved, the predicted time window, and the reason for concern | A vague red icon with no explanation |
| Data quality | Age, source status, uncertainty, or an explicit unknown or degraded label | False precision, or a confident line derived from stale data |
| Candidate path | Clearly marked as proposed; expiry and version; the constraints considered | Presenting a proposal as an ATC clearance or a guaranteed safe trajectory |
| Coordination status | Unconfirmed, acknowledged, timed out, or inconsistent | Calling a proposal agreed when acknowledgement messages are missing |
| System mode | Monitoring, advisory, degraded, or unavailable | Silently hiding failure or stale inputs |
| Existing alerts | No contradictory ANIRBAN command; display policy designed around approved procedures | Competing alert hierarchies, or suppressing ACAS/TCAS information |

## 9.3 The ghost path, demoted to a research visualization

The original concept described an augmented-reality windshield overlay in which the negotiated trajectory is drawn as a glowing line that the pilot follows by default, with divergence triggering alarms. That design is postponed. Version 0.1 uses a conventional advisory panel in which the proposed path is a clearly dashed line that must not resemble a flight-director command, and divergence from an advisory is an ordinary pilot act rather than an alarm condition.

The evidence in Section 9.1 predicts why. A confident overlay fed by uncertain data is the configuration that manufactures automation bias, and the compliance penalty for false alarms is measured rather than theoretical. Progression to an augmented-reality windshield is gated on controlled human-factors study results (stage V4, Section 12), because display and automation choices demonstrably change response behaviour. The first prototype displays advisory information only; separate evidence and approvals would be required before any active flight-control integration could be considered.

# 10 Preliminary Hazard Analysis, Safety Considerations, and Cybersecurity

## 10.1 Preliminary hazard register

Fourteen hazards were identified by merging the master requirement list with the supplied architecture document's HZ-01 to HZ-10 register. Severity is kept separate from probability throughout. No numeric risk scores are invented, and hazard classification authority remains a formal functional hazard assessment performed under a real certification basis by qualified personnel once the intended function and architecture are defined.

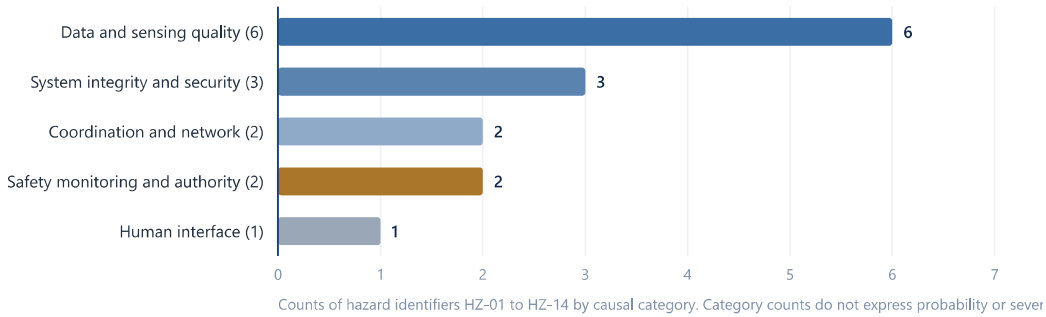



**Figure 8. Preliminary hazard register HZ-01 to HZ-14 by causal category**

Counts of the fourteen recorded hazard identifiers grouped by causal category: data and sensing quality 6; system integrity and security 3; coordination and network 2; safety monitoring and authority 2; human interface 1. Category totals agree with Table 15.

Source: Derived from the hazard register in Table 15 of this report; original identifiers from the supplied system-architecture design document (Rev 0.1).

Figure 8 shows where recorded hazards concentrate, and the distribution is informative about the system rather than about risk magnitude. Six of fourteen hazards arise from data and sensing quality — stale state, incorrect prediction, overconfidence, faulty fusion, missed conflicts and loss of navigation or timing. That concentration is a direct consequence of the architecture: ANIRBAN-1 adds no new actuation, so it cannot create hazards through control output, and nearly all of its failure potential lives in the quality of what it believes about the traffic picture. Category counts must not be read as probability, severity, or quantitative risk.

**Table 15. Preliminary hazard register (advisory-only prototype)**

| ID | Hazard / cause | Potential effect | Detection | Mitigation | Validation evidence needed |
|---|---|---|---|---|---|
| HZ-01 | Stale or invalid aircraft-state information | False or missed conflict prediction | Age checks, plausibility rules | Source and time validation; uncertainty and age display | Inject stale and plausible-but-wrong values; measure output gating |
| HZ-02 | Unsafe candidate trajectory passes the planner | Advisory misleads the operator | Independent re-check at L5 | All-track re-evaluation; constraint checks | Negative and boundary test cases |

| ID | Hazard / cause | Potential effect | Detection | Mitigation | Validation evidence needed |
|---|---|---|---|---|---|
| HZ-03 | Communication failure or network partition; delayed intent | Peers act on inconsistent plans | Link health, timeout state | No consensus dependency; local operation; proposal expiry | Partition fault-injection reports |
| HZ-04 | Malicious or misleading message (spoofed sender) | Coordinated advice built on a false track | Cross-source consistency, quality flags | Authentication plus explicit trust limits | Spoofing-like simulation |
| HZ-05 | Prediction overconfidence or incorrect prediction | Display implies unjustified certainty | Calibration metrics | Uncertainty visible; suppression when evidence is insufficient | Calibration study against ground truth |
| HZ-06 | Advisory contradicts a TCAS/ACAS alert or ATC instruction | Competing guidance confuses the crew | Resolution-advisory status via a permitted input | Suppress the conflicting proposal; document the authority hierarchy | Human-factors and procedure review |
| HZ-07 | Timing or processor failure | Stale advisory persists; an update is missed | Watchdogs, deadline monitors | Expiry, visible degraded state, controlled restart | Fault-injection and timing reports |
| HZ-08 | Faulty sensor fusion or false track | An object is missed or falsely declared | Confidence, modality cross-check | Bounded sensor scope; measured curves | Detection and false-alarm curves per target envelope |
| HZ-09 | Misleading cockpit presentation or ambiguity | Operator misinterprets a suggestion | Usability trials | Clear labelling; uncertainty and status exposure | Stage V4 study data |
| HZ-10 | Unauthorized data write path | Unvalidated data influences the system or the aircraft | Interface audit | Strict boundary; least privilege; no control outputs in the prototype | Architecture and code audit evidence |
| HZ-11 | Missed conflict | No advisory as separation erodes | Ground-truth scoring | Declared criterion with horizon sweep | Recall and false-negative statistics on labelled scenarios |
| HZ-12 | False alarm | Alert fatigue erodes compliance (a measured effect) | Scenario scoring | Threshold choice is a declared study parameter | Receiver-operating-characteristic trade curves across criteria |
| HZ-13 | Loss of navigation or timing information | The whole picture degrades simultaneously | Clock-quality status | Fail-closed suppression; degraded mode | GNSS-denied scenario tests |
| HZ-14 | Conflicting proposals between peers | Oscillation, or opposite manoeuvres proposed | Sequence and expiry correlation | Deterministic tie-break, or mutual suppression | Multi-peer adversarial scenario runs |

Two residual concerns are stated plainly. HZ-12: the false-alarm threshold cannot be chosen by engineering taste, because published alerting research shows that is exactly where crew tolerance is spent. HZ-14: the safest outcome under disagreement remains an open decision (Q5, Section 16), because two aircraft can each locally prefer a plan the other rejects, and the resolution rule must be written before multi-peer tests exist.

## 10.2 Cybersecurity and traffic-data integrity

The cooperative feeds on which ANIRBAN-1 depends are the most attackable links in the system, and this is measured, published ground rather than speculation. ADS-B was designed without cryptographic security mechanisms and is susceptible to message injection and sequencing manipulation. Published countermeasures trade cost against partial protection: a framework providing message authentication and integrity without depending on public-key cryptography at a bandwidth overhead below 50 per cent, and a planned aviation datalink security architecture combining national cryptographic algorithms for confidentiality and integrity with TESLA-based authentication of control data at an average byte overhead below 10 per cent. The operational threat is documented: national security authorities have reported foreign intelligence penetration of connected drone inspection platforms, with exploited weaknesses catalogued as unpatched vulnerabilities and weak credentials, ambiguous permission boundaries, unencrypted data transmission and device-firmware backdoors. Regulatory response is moving from guidance toward mandate, with a mandatory national standard on civil-UAV data-link cybersecurity in drafting that covers identity authentication, anti-forgery and anti-tampering, secure updates and anti-replay protection.

**Table 16. Threat, design control, and what each control does not prove**

| Threat or fault | Design control | What the control does not prove |
|---|---|---|
| Spoofed sender or forged message | Identity and key management; message authentication where suitable | Does not prove that the sender's sensors or reported intent are truthful |
| Replay or stale message | Sequence numbers, timestamps, expiry, anti-replay checks, clock-quality status | Does not eliminate all time-source failure or compromised-sender risk |
| Corrupted or malformed data | Strict schema validation, range checks, version negotiation, fail-closed handling | Plausible values can still be wrong |
| Network partition or jamming | Timeouts, link health, bounded retries, degraded mode, local prediction | Cannot guarantee message availability when the link is absent |
| Compromised process | Process isolation, least privilege, signed builds and updates, logging, watchdogs | Requires a broader system threat model and verification |
| Unsafe configuration or update | Versioned configuration, integrity checks, controlled installation, rollback | Does not replace authorized maintenance and configuration management |

Privilege separation is the architectural counterpart: the network gateway, prediction service, user interface and maintenance or log-export path run under separate privileges, and no ground analytics service or internet-reachable component may write an unchecked trajectory into the advisory path.

## 10.3 Authority boundaries and operational limitations

The pilot and air traffic control retain authority. ANIRBAN-1 output is explicitly advisory and does not change the legal or operational authority structure. Liability questions remain open and are noted rather than resolved: in concept the crew retains final authority while the system negotiates with other aircraft, and any incident investigation would examine algorithmic decisions, which is why validation evidence and complete logging (SYS-010) are treated as first-class deliverables.

Operational limitations are stated without softening. Full benefit requires equipage density, so legacy unequipped traffic does not benefit. Non-cooperative obstacle detection is limited by weather, lighting, geometry and processing. GNSS-denied conditions degrade intent sharing. Fallback to existing ATC separation and to TCAS/ACAS is therefore mandatory, not optional.

## 11 Applicable Standards and Potential Certification Pathway

The standard inventory in the supplied documents required correction before use, because several citations name superseded revisions or imply a compliance relationship that does not exist. The system-development guidance cited as ARP4754A has been succeeded in the development-process chain by ARP4754B, which is aligned with ARP4761A for safety assessment; ARP4761A provides the hazard-identification and risk-assessment methods (functional, preliminary system and system safety assessment). DO-178C remains the primary airborne-software assurance means with development assurance levels A to E, and DO-254 is the matching process for complex electronic hardware. The functional relationship matters more than the catalogue: ARP4754-class guidance ensures the system is developed correctly, ARP4761-class guidance ensures the correct system is developed, and the outputs of the development process — safety requirements and assurance-level allocation — drive the software and hardware verification activities beneath them.

**Table 17. Standards and guidance mapped to ANIRBAN-1 activities**

| Domain | Guidance (current designation) | Role for this programme | Status caveat |
|---|---|---|---|
| Airborne software | DO-178C / ED-12C | Requirements-to-code verification rigour for the deterministic core, if an assurance track is pursued | Recognized by FAA advisory circular AC 20-115D as an acceptable means, not the only means; applicability is project-specific |
| Complex electronic hardware | DO-254 / ED-80 | Applies to any FPGA or ASIC developed for radios or gate arrays in a future H2 or H3 build | Only relevant if complex hardware is actually developed |
| System development | ARP4754A, succeeded by **ARP4754B** | Development process, assurance-level allocation, requirements linkage | A process framework, not a certificate |
| Safety assessment | ARP4761, succeeded by **ARP4761A** | Functional, preliminary system and system safety assessment methods behind the hazard register in Table 15 | A formal assessment requires a defined intended function and aircraft |
| Environmental and EMC | DO-160-class test categories | Qualification of airborne-equipment candidates | Applies to H3-class design studies only |
| Cybersecurity | DO-326A / DO-356A-class process; emerging national datalink security standards | Threat model, security requirements and verification | Rule-creation status varies by authority; current mandates must be verified per jurisdiction |
| Human factors | ARP5365A cockpit traffic display interface standard, plus authority human-factors policy | Display and alerting design evidence for stage V4 | The standard covers traffic displays; it does not address augmented-reality overlays specifically |

| Domain | Guidance (current designation) | Role for this programme | Status caveat |
|---|---|---|---|
| Modification approval | FAA and EASA supplemental type certificate routes and equivalents | Relevant only after a real aircraft modification is proposed | Listing an approval route is not approval; no timeline is asserted |

> **Two statements that must travel with any excerpt of this section.** First, no standard named in this report means ANIRBAN-1 currently complies with that standard. Second, certification is not a final paperwork phase: it shapes system requirements, architectural independence, test evidence, configuration control and human-factors work from the beginning. Assurance levels are not assigned in this document; they are outputs of a formal assessment under a defined certification basis, not inputs to a concept.

The credible pathway from software simulation to hardware-in-the-loop testing, controlled research equipment and potential aircraft integration is staged in Section 13. Stage V5 exists specifically to convert standards awareness into an aircraft-specific feasibility decision made with a manufacturer or design organization and the applicable authority; stage V6 admits trials only under proper authorization, initially passive and record-only.

# 12 Validation Methodology, Baselines, Metrics, and Acceptance Criteria

## 12.1 Validation stages

No live aircraft installation is needed to establish the first meaningful technical result. The programme advances through declared stages, each with exit evidence rather than a date.

**Table 18. Validation stages V0–V6 with exit evidence**

| Stage | Environment | Test focus | Exit evidence |
|---|---|---|---|
| V0 Model review | Requirements and code review | System boundary, definitions, assumptions, data contracts, threat model | Approved review checklist; open issues recorded |
| V1 Software simulation | Multi-aircraft simulator or custom scenario engine | Crossing, converging and overtaking geometries; intent changes; uncertainty; multiple traffic | Reproducible scenario set; baseline metrics |
| V2 Network simulation | Virtual network with injected loss, latency, reordering, duplication, partition and replay-like cases | Message semantics, timeouts, inconsistent intent, degraded state | Fault-injection results; safe output-gating evidence |
| V3 Hardware-in-the-loop | Bench computer with simulated I/O; optional lab sensors or radios | Timing, restarts, resource exhaustion, interface fault behaviour | Timing and recovery reports; confirmation that no unauthorized radio-frequency transmission occurred |

| Stage | Environment | Test focus | Exit evidence |
|---|---|---|---|
| V4 Human factors | Mock display and controlled operator study | Comprehension, response, workload, alert interpretation, trust calibration, rejection of incorrect advice | Study protocol, ethics handling where applicable, data and analysis |
| V5 Aircraft feasibility study | Aircraft-specific design and safety review before installation | Interfaces, power, EMC, environmental conditions, airworthiness and certification route | Qualified engineering decision and approval plan |
| V6 Approved ground and flight trials | Only under proper authorization and an approved test plan | Incremental passive and record-only operation first, then carefully bounded advisory trials | Formal test reports; authority and manufacturer approvals |

Retained from the supplied architecture document. BlueSky is one candidate open-source air-traffic simulation starting point for stages V1 and V2.

## 12.2 Baselines and the ablation discipline

Three baselines define what "improvement" would mean. **Baseline A** is constant-velocity extrapolation with a geometric proximity test. **Baseline B** is a DAIDALUS-style detect-and-avoid implementation: NASA's reference logic provides a well-clear definition, look-ahead prediction of violation intervals, conflict and recovery guidance bands, and graduated alerting, and is openly available with prototype implementations and formal specifications. **Baseline C** is a published four-dimensional trajectory conflict-detection and resolution method whose documentation permits reproduction; polynomial-based distributed resolution minimizing deviation from original trajectories is the candidate recorded in the supplied comparison.

The ANIRBAN variant is intent sharing plus uncertainty handling plus fault-aware coordination. **Ablation runs must disable one feature at a time.** Without ablation, a measured improvement cannot be attributed to any component — which is precisely the standard that concept claims of this kind have so far failed to meet.

## 12.3 Metric families

**Table 19. Metric families to be defined before testing**

| Metric family | Candidate measures |
|---|---|
| Detection | Conflict detection rate, missed-conflict rate, false-alarm rate, warning lead-time distribution |
| Prediction | Position prediction error by horizon, calibration of uncertainty, behaviour under intent change |
| Resolution | Feasibility rate, modelled constraint violations, path deviation, manoeuvre feasibility, oscillation and repeated replanning |
| Network | Message age, end-to-end delay distribution, packet loss, stale-message rejection, recovery after partition |
| Runtime | Processing-time distribution, deadline misses, processor and memory use, restart time, event-log completeness |

| Metric family | Candidate measures |
|---|---|
| Human factors | Time to understand an advisory, correct manoeuvre selection, workload, trust calibration, and rates of inappropriate acceptance or rejection |

Acceptance thresholds are set before testing, derived from stated research aims and engineering constraints. Arbitrary percentages are not borrowed, and no flight-safety assurance is claimed from small simulation studies.

### 12.4 Research questions that define success

1. Does sharing a planned trajectory improve prediction lead time over current-state-only traffic, measured against Baselines A to C?
2. Under what message-loss conditions does distributed coordination stop adding value?
3. Does uncertainty-aware sensing reduce missed detections without creating unacceptable false alerts, judged by measured detection and false-alarm distributions?
4. Does the interface help operators correctly distinguish safe from unsafe proposals?
5. What measurable benefit survives after accounting for false alarms and for legacy or unequipped aircraft?

## 13 Development Roadmap and Evidence Gates

The original concept committed to a fixed five-to-seven year schedule with calendar milestones. That commitment is replaced by phase gates whose exit criteria are evidence, because certification duration depends on the certification basis, the authority and the programme scope, none of which can be asserted from a concept document. Figure 9 shows the illustrative sequencing; the gates govern.

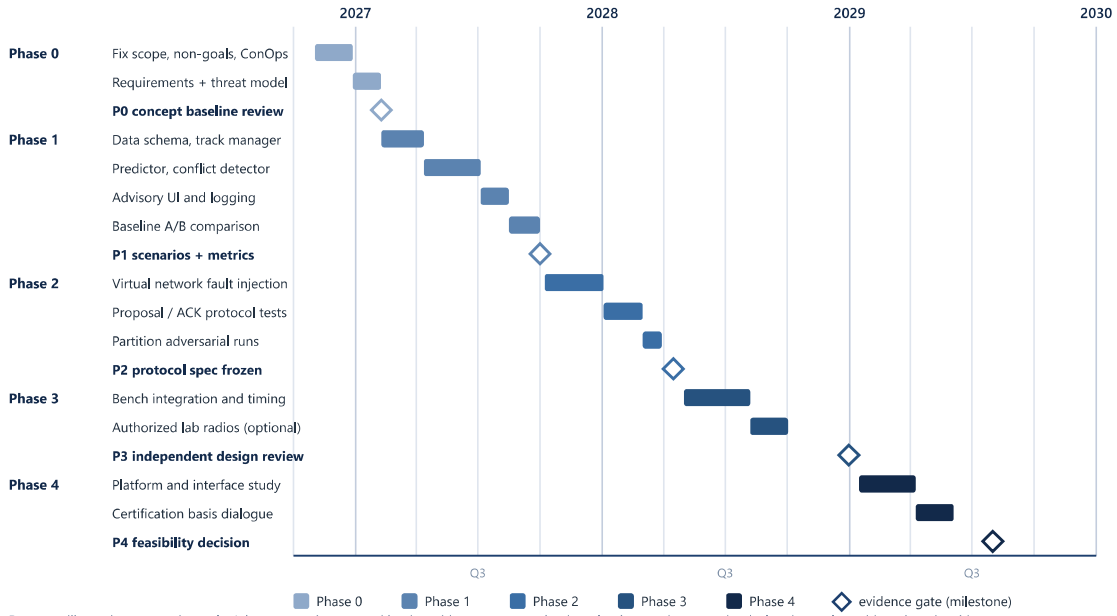



**Figure 9. Development roadmap: illustrative sequencing with mandatory evidence gates**

Five phases from concept baseline to research aviation study. Bars show illustrative activity windows; open diamonds mark evidence gates (P0 concept baseline review, P1 repeatable scenarios and baseline metrics, P2 protocol specification frozen, P3 independent design review, P4 feasibility decision). Dates are planning assumptions, not commitments.

Source: Phase content from the supplied system-architecture design document (Rev 0.1). Calendar dates are illustrative planning assumptions introduced in this revision and are not derived from any source estimate.

**Reading the schedule correctly.** The visual schedule is subordinate to the technical decision gates. Assigning a date does not make aircraft integration, operational trials or certification automatic: Phase 4 cannot begin without a selected platform and an engaged authority, and no trial in Phase 5 or beyond may proceed without the approvals its gate requires. A phase may not be declared complete without its exit evidence, regardless of elapsed time.

**Table 20. Phase gates: scope and exit evidence**

| Phase | Scope | Exit gate (evidence, not calendar) |
|---|---|---|
| Phase 0 — Concept baseline | Intended use, airspace and user context, source interfaces, assumptions, non-goals, hazards, benchmark plan | Concept of operations; system boundary; initial requirements; literature evidence map |
| Phase 1 — Simulation MVP | Data schema, track manager, predictor, conflict detector, advisory interface, logging | Repeatable scenarios; baseline metrics; no radio or aircraft modification |
| Phase 2 — Robustness and network | Virtual network; proposal and acknowledgement protocol; timeouts; authentication simulation; partition cases | Protocol specification; threat model; failure-state evidence |
| Phase 3 — HIL prototype | Bench computer; simulated avionics I/O; optional controlled lab sensors or radios | Interface test report; timing and fault report; independent design review |

| Phase | Scope | Exit gate (evidence, not calendar) |
|---|---|---|
| Phase 4 — Research aviation study | Select platform and function; engage a manufacturer or design organization and the authority | Aircraft-specific feasibility; certification-basis discussion; test plan |
| Phase 5 — Authorized trials | Only after required approvals; passive and record-only first, then bounded advisory operation | Conformity and test evidence required by the approved programme |
| Phase 6 — Operationalization study | Procedures, training, maintenance, interoperability and continuing airworthiness — only if evidence justifies it | Certification and operational evidence; no approval assumed |

Phases 5 and 6 lie beyond the roadmap drawn in Figure 9 and are entered only on the strength of Phase 4 evidence.

### 13.1 Cost and retrofit considerations (qualitative)

**Table 21. Cost and complexity categories by hardware option**

| Cost category | H1 (bench) | H2 (research island) | H3 (airborne-class) |
|---|---|---|---|
| Hardware | Software development time only | Workstation plus adapter and watchdog units | Qualified line-replaceable-unit development and testing |
| Integration | Simulated I/O | Laboratory interfaces; optional authorized radios | Airframe-specific engineering, EMC and environmental qualification |
| Assurance | None required (research activity) | Internal design review | Full ARP4754B and ARP4761A-class process, with DO-178C rigour per assigned assurance level |
| Approval | Not applicable | Not applicable | Supplemental-type-certificate-class modification approval per aircraft |

The original concept report's per-aircraft dollar bands, power-draw figures, fuel-burn impact estimate and itemized vendor pricing are removed as unverified. Restating them would require vendor quotations under a defined installation scope and airframe-specific weight and power analysis, neither of which exists. Adoption incentives such as equipage mandates are policy questions noted here but not forecast.

## 14 Discussion, Research Gap, Proposed Contributions, and Limitations

### 14.1 Mapping the occupied territory

Layering ANIRBAN-1 against the evidence base produces an honest overlap map. Trajectory mathematics: occupied. Conflict screening and detection research, including space-time reachability treatments of motion and pilot-intent uncertainty, and polynomial-based distributed resolution: occupied. Manoeuvre optimization including learning-based methods: occupied. Alerting and guidance logic with formal verification: occupied by DAIDALUS, and being operationally superseded by the ACAS X family. Intent sharing to ground services and between air-traffic systems: under active deployment in national TBO programmes. Air-to-air broadcast of current state: fielded through ADS-B. Air-to-air ad hoc networking services: standardized in draft form.

## 14.2 The surviving research gap

What remains genuinely open is a coherent, testable specification of fault-aware peer coordination: message expiry and staleness semantics; deterministic disagreement and tie-break outcomes; explicit unconfirmed states when peers do not answer; partition behaviour that preserves local detection; and an advisory interface that exposes all of this to a human. These are protocol-and-evidence problems rather than algorithm problems, which is why Sections 7 and 9 carry the novelty burden in this revision.

## 14.3 Candidate contributions

**Table 22. Candidate contributions stated as hypotheses, with the evidence each requires**

| ID | Hypothesised contribution | Evidence required to claim it | Current status |
|---|---|---|---|
| C1 | A fault-aware state and intent exchange schema with uncertainty, time validity and explicit degradation rules | Schema specification; stage V2 results showing correct expiry and staleness behaviour versus an ADS-B-style state-only feed | Specified conceptually; untested |
| C2 | A coordination protocol that remains predictable under disagreement and message loss | Deterministic tie-break specification; adversarial multi-peer runs at stages V1 and V2; no undefined states in the state machine | Design drafted; the disagreement outcome is still an open decision (Q5) |
| C3 | A combined reproducible evaluation of trajectory sharing, sensing uncertainty and advisory quality in one simulation and hardware-in-the-loop environment | Open scenario corpus; baseline comparison table; ablation attribution | Environment plan exists; nothing executed |
| C4 | An advisory presentation that exposes uncertainty and data freshness rather than a confident path | Stage V4 study with calibrated-trust and correct-rejection measures, referenced against published ACAS Xu display-response findings | Design principles set; study not run |

A credible novelty claim requires a documented literature search and baseline comparisons. The literature already contains multi-aircraft four-dimensional conflict detection and resolution, optical detect-and-avoid, collision avoidance optimization, and well-clear algorithms, so these four items are the defensible residue and are stated as hypotheses contingent on the paired evidence.

## 14.4 Limitations of this report

- **No measurements exist.** Every ANIRBAN-1 performance characteristic in this report is a design target or a hypothesis. Nothing has been flown, and no simulation campaign has been run.
- **Unverified source numbers are flagged, not corrected.** Where the supplied documents conflict with each other — sensor ranges, alert lead times, uptime targets — both values are recorded and the item is marked for author or measurement confirmation.
- **The project name has no public footprint.** Searches across academic and government platforms returned only unrelated personal-name matches for the project designation, which is consistent with a private research project and means no external corroboration of its claims is available.
- **Standards status is as-reported.** Revision designations and advisory-circular status were checked against the supplied references; a formal applicability determination belongs to the certification authority for a specific project.
- **Cost and schedule are qualitative.** No verified vendor, weight, power or approval-duration data was available, so none is asserted.

# 15 Decisions Made During Revision of the Original Concept

## 15.1 Disposition of the original functional elements

Twelve functional elements of the original concept were dispositioned. Figure 10 summarizes the outcome and Table 23 records the reasoning element by element. The changes matter because they define what Version 0.1 is allowed to claim: retaining intent sharing and the independent safety monitor while removing consensus and postponing control execution, augmented-reality overlays, universal sensing and 6G dependency produces a smaller programme whose research question can actually be answered with the resources a research team has.

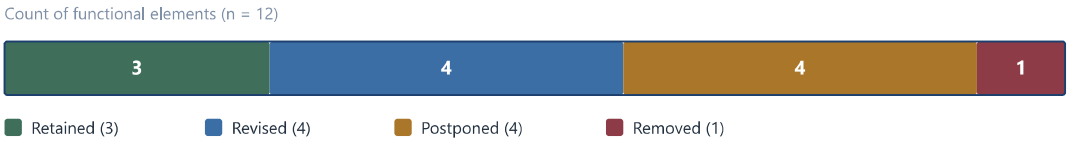



**Figure 10. Disposition of the twelve functional elements of the original concept**

Three elements retained, four revised, four postponed behind evidence gates, and one removed from the time-critical safety loop. Segment widths are proportional to counts and total twelve.

Source: Derived from the element-by-element disposition recorded in Table 23 of this report.

**Table 23. Element-by-element disposition and rationale**

| Original element | Disposition | Reason |
|---|---|---|
| Aircraft as connected nodes in an air-to-air network | Revised | Retain the networking concept but anchor it to IEEE 1920.1-class airborne ad hoc networking rather than an invented transport layer |
| Aircraft-state and planned-trajectory sharing | Retained | The core testable hypothesis; complements ADS-B state broadcast, which does not carry a negotiated future four-dimensional path |
| Forward-looking conflict prediction | Retained | Must be evaluated against DAIDALUS-style well-clear look-ahead rather than asserted |
| Multi-aircraft coordination and resolution proposals | Revised | Restricted to timeout-bounded proposal and acknowledgement exchange with explicit disagreement handling |
| Blockchain or PBFT-style consensus in the safety loop | Removed | Adds latency and failure modes; cannot establish sensor truth; unnecessary for local detection |
| Automatic trajectory execution on the autopilot | Postponed | Requires a separate flight-control integration, human-factors and certification case before any control path exists |
| Augmented-reality windshield ghost path with auto-follow default | Postponed | Start from a conventional advisory display; published alerting and situational-awareness findings require evidence first |
| LiDAR, optical and radar non-cooperative sensing | Postponed as a separate study | Retained as an offline research module; detection claims need a measured target envelope |

| Original element | Disposition | Reason |
|---|---|---|
| Dedicated 6G aviation mesh | Postponed | Requirements work ongoing with commercial framing around 2030 |
| Independent safety monitor | Retained | A strong element of the original concept; formalized as layer L5 with declared independence limits |
| Ground and ATC integration | Revised | ATC oversight retained; any operational-concept change is institutional and needs controller human-factors evaluation |
| Certification appendix asserting DAL A and an STC route | Revised | Converted from asserted compliance mapping into a pathway analysis tied to each standard's actual role |

Category totals agree with Figure 10: three retained, four revised, four postponed, one removed.

## 15.2 Claims requiring confirmation, and numbers removed

**Table 24. Unresolved claims and removed figures**

| Claim or item | Source | Gap type | Disposition and confirmation route |
|---|---|---|---|
| TCAS resolution advisory issued roughly 20–25 seconds before collision | Original concept report | External, verifiable | Retained as an unverified original claim. Public material describes a roughly five-second pilot response window on a resolution advisory; the 20–25 second lead figure needs its own citation before use |
| Communication uptime above 99.999 per cent and mesh latency around 100 ms | Original concept report | External, engineering | Removed as an unverified target. Requires link-level simulation and authorized radio trials |
| Conflict resolution success in X per cent of scenarios; no near-mid-air-collision events in thousands of flight hours | Original concept report | Placeholder, internal | Retained as placeholders. The team must set numeric acceptance thresholds before testing |
| Development assurance level A assigned to collision-avoidance functions | Original concept report | Regulatory | Removed. Assurance levels are outputs of a formal assessment under a defined certification basis, not concept inputs |
| Per-aircraft hardware cost of roughly USD 100,000 to 500,000, with itemized sensor, compute and radio prices | Original concept report | Unverified, external | Removed. Requires current vendor quotations under a defined installation scope |
| Five-to-seven-year certification timeline | Original concept report | Unverified, regulatory | Replaced by evidence gates (Section 13). Duration depends on basis, authority and scope |
| Fuel-burn impact below 1 per cent after noise-mitigation measures | Original concept report | Unverified, engineering | Removed. Requires airframe-specific weight and power analysis |
| Sensor range figures inconsistent between the concept body and its own comparison table | Original concept report | Internal inconsistency | Both values recorded in Table 7; flagged for measurement rather than averaged |

| Claim or item | Source | Gap type | Disposition and confirmation route |
|---|---|---|---|
| Contribution statements containing unfinished area placeholders | Contribution note | Internal, authorship | Reproduced verbatim in Section 19 with the placeholders visibly flagged as author-action items |

### 15.3 Presentation defects corrected in this edition

- The standalone opening chart gallery was removed. Each visualization now sits beside the argument it supports and is cited explicitly in the text (Figures 1 to 10).
- The duplicated glossary and contributor block at the end of the exported PDF was eliminated; each appears once, as Sections 18 and 19.
- Tables that collapsed into single merged cells on PDF export were rebuilt with proper header groups and page-break behaviour, so headers repeat across pages.
- The radar update interval, previously drawn as a single bar at the top of its reported range, is now drawn as a hatched band spanning the reported 4–12 second range on a logarithmic axis with the scale choice explained (Figure 6).
- Generic series names were replaced with descriptive labels carrying units; redundant legends were removed; estimates, projections, ranges and programme-reported claims are labelled as such in every caption.
- Charts are now inline SVG rather than canvas rendered from remote scripts, so figures cannot appear blank or malformed in the PDF edition.

## 16 Conclusions and Next Steps

### 16.1 Conclusion

ANIRBAN-1 is worth building precisely because the systems around it already work. Probabilistic multi-directional collision avoidance, open well-clear alerting logic with formal specification, verified four-dimensional trajectory sharing, and standardized air-to-air ad hoc networking all exist. The open frontier is not smarter avoidance and not a new network. It is whether aircraft can share intent and coordinate under fault, uncertainty and disagreement without ever making their existing safety depend on that cooperation. That question is answerable with bench computers, simulators and honest baselines, and the scope defined in this report is sized to answer it.

Until the ablation baselines of Section 12 have been run, the correct statement of ANIRBAN-1 status is that it is a well-posed research hypothesis with a credible, low-cost test path — not a demonstrated safety improvement, and not a replacement for any existing system.

### 16.2 Open decisions gating implementation

**Table 25. Open questions, why they matter, and what triggers each decision**

| # | Open question | Why it matters | Decision trigger |
|---|---|---|---|
| Q1 | First target: airliner, research aircraft, or unmanned aircraft | Changes the traffic model, interfaces, target environment and airworthiness path | Before selecting real equipment or a certification strategy |

| # | Open question | Why it matters | Decision trigger |
|---|---|---|---|
| Q2 | Which separation or well-clear criterion governs | All conflict results depend on the chosen criterion and operating domain | Before creating test ground truth |
| Q3 | Which traffic and intent standard is exchanged | A message format without an interoperable service is not deployable | Before integration beyond simulation |
| Q4 | How uncertainty and unknown data are represented | Determines false negatives, false alarms and suppression behaviour | Before baseline comparison |
| Q5 | Safest coordination outcome under disagreement | Multi-aircraft interactions can oscillate or produce contradictory advice | Before enabling peer proposal tests |
| Q6 | Which sensor target envelope is in scope | Prevents unsupported universal-detection claims | Before selecting any LiDAR, camera or radar module |
| Q7 | Which authority and certification basis applies | Determines evidence, rigour and allowable operation | Before any aircraft modification |

### 16.3 Immediate next steps (twelve-month action set)

1. Freeze the Version 0.1 requirements (Table 4) and the message schema (Table 9).
2. Stand up the Baseline A and B simulation harness with the ablation switchboard described in Section 12.2.
3. Run the stage V2 fault-injection catalogue against the deterministic coordination protocol and record every state transition against Table 3.
4. Document the threat model including the explicit authentication-limits statement from Section 7.2.
5. Draft the stage V4 human-factors protocol against cockpit traffic display interface requirements.
6. Open decision records for Q1 to Q7 and complete the contributor statements in Section 19.

## 17 References

Supplied project materials are treated as project context, not as verified evidence of system performance. External sources were checked for status and role; where a claim could not be established it is marked for author verification in Table 24 rather than presented as confirmed.

#### Supplied project documents

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**Verification note.** A public-name check found no aviation-system usage of the project designation across academic and government platforms; only unrelated personal-name matches were returned. This is consistent with a private research project and means no external corroboration of project-specific claims is available. All ANIRBAN-1 features and expected benefits remain proposals until they are defined and measured in a controlled study.

## 18 Glossary

**ACAS / TCAS**
Airborne Collision Avoidance System / Traffic Collision Avoidance System.

**ADS-B**
Automatic Dependent Surveillance–Broadcast; a cooperative traffic-state source where aircraft are equipped and coverage is available.

**Advisory-only**
Displays information and recommendations; outputs no flight-control command.

**AFDX**
Avionics Full-Duplex Switched Ethernet (ARINC 664 Part 7); an aircraft-specific networking possibility, not a universal interface.

**4DT**
Four-dimensional trajectory: latitude, longitude, altitude and time.

**ATC**
Air Traffic Control.

**CD&R**
Conflict Detection and Resolution.

**CDTI**
Cockpit Display of Traffic Information.

**DAL**
Development Assurance Level (A to E), an output of formal safety assessment.

**DAIDALUS**
NASA Detect and Avoid Alerting Logic for Unmanned Systems; an open reference implementation of well-clear alerting and guidance bands.

**Degraded state**
The system remains available for limited purposes but cannot support its normal confidence level; advice may be suppressed.

**FMS**
Flight Management System.

**FHA / PSSA / SSA**
Functional Hazard Assessment / Preliminary System Safety Assessment / System Safety Assessment (ARP4761-class methods).

**HIL**
Hardware-in-the-loop: prototype hardware tested against simulated inputs, plant and traffic.

**ICD**

Interface Control Document: message definitions, units, rates, access rights and failure semantics.

**Intent**
A stated intended future trajectory; a claim about planned behaviour, not proof of what the aircraft will do.

**MVP / Version 0.1**
Minimum viable prototype; here, a simulated, advisory-only research implementation.

**RA / TA**
Resolution Advisory / Traffic Advisory.

**STC**
Supplemental Type Certificate; an approval route for a modification to an aeronautical product.

**TBO**
Trajectory-Based Operations.

**Unconfirmed state**
A declared coordination outcome when a peer does not respond by the deadline; never treated as acceptance.

**Well-clear / conflict criterion**
The declared mathematical threshold used to label potentially unsafe proximity; it must be defined for the research domain.

## 19 Contributor Statements

**Abu Kayes Huda:** Conceived the entire concept and designed the full system. Developed the IoT mesh architecture, node communication, real-time data routing, redundancy/failover, autonomous aircraft safety logic, and system integration.

**Jaidul Islam Alvi:** Participated in early discussions and provided feedback on [Idea and System Architecture] — Reviewed draft material.

**MD Sakib Hasan:** Participated in discussions and provided input on [System Architecture and Core Concept] — Reviewed the final report/presentation.

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This revision was prepared by an aviation systems engineering and technical editing review of the supplied project materials. Corrections to the original concept are recorded in Section 15 with their rationale. Placeholders and unverified figures are deliberately left unresolved rather than completed by inference.

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